



The Spaghetti Engine
VISUAL SYSTEM MODELLING WORKSPACE

The Spaghetti Engine

Development Log

Fuzzy Cognitive Map modelling tool

Field	Value
Version	0.4.0 (prototype)
Log entry date	26 September 2026
Status	Built and verified locally; not yet deployed. Version 0.3.0 remains the live release until this one is pushed.
Live URL	spaghetti-engine.netlify.app (Netlify, auto-deployed from the GitHub main branch)
Source repository	github.com/aPaarl/Spaghetti-engine (release commit to be recorded when 0.4.0 is pushed to main)
Previous version	0.3.0, 15 September 2026 (see Spaghetti_Engine_Dev_Log_v0.3.0.docx)
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1. Purpose of This Log

This Development Log records the changes from version 0.3.0 (15 September 2026) to version 0.4.0 of The Spaghetti Engine. Version 0.4.0 is a feature and usability release. It adds a tenth tab, Monte Carlo, which tests how far the results depend on the uncertainty in the relationship weights; it lets every simulation run without a squashing function, so the raw dynamics of a map can be explored; it groups the tabs by what the user is doing (model building, state and scenario analysis, sensitivity analyses, transition analyses); it reworks the Network tab for larger models and for direct manipulation of relationships; and it adds one system for publication-quality figures across the analysis tabs. The gain measure of the Transition Pathways tab is called total outcome gain, the term of the case study of the accompanying paper. The peat meadow case study is added as a third example model. A new logo and brand identity are introduced, together with an About & documentation tab that offers the documentation as PDF downloads.

The log documents what was added, the implementation choices that go beyond the established literature or that testing forced, the defects found during development and how they were resolved, how the work was verified, and the deployment status. It is a technical companion to the User Manual and the Methodological Annex, both updated to version 0.4.0 alongside this log (Section 10).

2. Summary of Version 0.4.0

Area	Change
Monte Carlo (new tab)	Re-runs the model hundreds or thousands of times with every unlocked relationship weight (and optionally every starting value) nudged by random noise, and reports per concept the nominal value, mean, median, standard deviation, percentiles, minimum and maximum, with histograms and a CSV export. Seeded, so a run can be repeated exactly.
No squashing function	A fourth transfer-function choice, 'linear', next to hyperbolic tangent, sigmoid, and trivalent, on Baseline Equilibrium and in the shared simulation settings. No bound and no clamp; a run that grows past 1000 is stopped and reported as diverged.
Tab groups	The ten tabs are arranged in four groups: Model building, State and scenario analysis, Sensitivity analyses, and Transition analyses.
Network tab	A larger canvas with a collapsible Inspector; hover isolation of a concept and its connections; search for a concept; a switch that shows relationships by arrow color and width only; drag-to-bend and drag-to-reconnect for relationships, with snapping to a visible anchor point; relationship arrows that never sit above concept boxes; smoother dragging and hovering; a stricter undo history.
Publication-quality figures	One figure panel (size presets in millimeters, DPI, fonts, axis titles, palettes including a color-blind-safe and a grayscale one, legend position, line and marker settings, which series to show and in what order) and one exporter (PNG with embedded DPI, SVG, single-page PDF), applied to ten charts on four tabs.
Transition Point Analysis	The results table sorts on every column and gains Range and Efficiency columns.

Area	Change
Sensitivity Analysis	The impact ranking and the elasticity metrics are proper bordered tables with right-aligned numbers, and each has its own CSV export.
Transition Pathways	The gain measure is called total outcome gain (TOG) in every label, chart axis, and export. The selected pathway is labelled on the effort-gain plane; the outcome trajectory and the benefit accrual chart (now a filled area) follow the figure panel.
Example model	The peat meadow nexus (20 concepts, 50 relationships, five modules, seven scenarios) is added as a third preloaded example; it loads with a 300-iteration limit.
Branding and About tab	A new logo (an S-shaped strand of spaghetti through a causal map, with a cog at its center), a palette of cobalt, navy, and a citrus accent ('Cobalt and citrus') applied to the whole interface, and a favicon. A new About & documentation tab describes the tool, who it is for, and why it was made, offers the three documents as PDF downloads, and gives the contact details.
Reliability	A troubleshooting pass over the release found and fixed six defects, among them stale results after replacing a model, a Monte Carlo run that could not be repeated in asynchronous mode, an off-by-one in the reported iteration cap, and missing warnings for runaway runs; the full list of defects found during development is in Section 6.
Version	Raised from 0.3.0 to 0.4.0 in package.json and package-lock.json.

3. New Features

3.1 The Monte Carlo tab

- **Method.** runMonteCarlo is a JavaScript generator, run in slices of about 40 ms like the pathway analysis, so the page stays responsive, a progress bar shows how far it is, and Cancel stops it. For each run it draws a new weight for every relationship not pinned by the scenario, uniformly from the recorded weight plus or minus the chosen spread, clipped to -1..+1 and rounded to two decimals; optionally does the same for every starting value not pinned by the scenario; and simulates with the ordinary engine. Locked concepts, overridden weights, and overridden starting values are never perturbed.
- **Randomness.** mulberry32, a small seeded 32-bit generator, so that the same model, settings, and seed reproduce the same distribution exactly. In asynchronous mode the random update order has its own seeded stream (Section 4), so the seed also fixes the order.
- **Settings.** Scenario to test, number of runs (100, 250, 500, 1,000, or 2,000; default 500), weight uncertainty (0 to 1, default 0.1), starting-value uncertainty (default 0), and the seed, with a button that draws a new one.
- **Results.** summarizeMonteCarlo reports, per concept, the nominal value (the unperturbed run), mean, median, standard deviation, minimum, maximum, and the 5th, 25th, 75th, and 95th percentiles; the number and share of runs that converged and, without a squashing function, that ran away. The header states the settings the run used. The three concepts with the largest spread are charted as histograms on completion, and any concept can be added or removed. The summary can be exported as CSV.
- **Conventions.** The usual amber banner appears when the model changes after a run; results are not saved; leaving the tab cancels a run in progress. The tab is not part of the Export Analysis Workbook and does not use the figure panel (Section 9).

3.2 Running without a squashing function

- **What it is.** The update $A(t+1) = A(t) + W A(t)$ (relative rule) or $W A(t)$ (absolute rule), with the identity as transfer function. It is offered as a third button on Baseline Equilibrium ('No squashing (linear)') and as 'None (linear, unbounded)' in Scenarios & Simulation, Advanced options, which sets it for every simulation in the app (Sensitivity, Transition Point, Monte Carlo, and Pathways included). The steepness control is hidden where it has no meaning.
- **No clamp, and a runaway guard.** Activation is not clamped to $-1..+1$ either, since a clamp would be a hidden squashing function. A linear map can grow without limit, so a run is stopped once any concept passes 1000 in absolute value (or becomes non-finite) and is flagged as diverged. Baseline Equilibrium then reports a Divergence status with an explanation; Scenarios & Simulation counts the runs that diverged; the chart axes rescale instead of staying at $-1..+1$.
- **Consequences for other analyses.** The Influence and Sensitivity scores need a bounded scale and keep using tanh when squashing is off, which the method panel states. The pathway engine mirrors the runaway rule; its robustness test replaces the steepness variants by a comparison with tanh. Method panels and exports record the transfer function as 'none (linear, unbounded)'.
- **Example.** On the peat meadow example the baseline settles under tanh in 16 iterations, but without squashing it runs away and is stopped at iteration 13 with the largest activation about 1,700 (the reinforcing loops of the model amplify themselves on every pass, which the squashing function normally hides).

3.3 Tab groups

A row of four group buttons sits above the tabs; the tabs of the active group appear beneath it. Model building holds Model editor, Network, Influence Metrics, and Influence Routes (the tabs a participatory session uses). State and scenario analysis holds Baseline Equilibrium and Scenarios & Simulation. Sensitivity analyses holds Sensitivity Analysis and Monte Carlo. Transition analyses holds Transition Point Analysis and Transition Pathways. Every tab keeps its previous identifier, so links between tabs still work and switch the group automatically; a group remembers the tab last used in it. On narrow screens the group buttons wrap instead of scrolling.

3.4 The Network tab

- **More room, less clutter.** The canvas is as tall as the window allows instead of a fixed 700 pixels; the Collapse panel button hides the Inspector so the map fills the width; the toolbar has dividers between groups of controls and a short information icon in place of the always-visible hint.
- **Isolating a concept.** Hovering a concept keeps it and its direct connections at full strength and fades everything else; the base opacity of relationships adapts to how dense the map is. A search box finds a concept by name, unfolds its module if it is folded, selects it, and centers the map on it.
- **Labels.** The Labels switch hides the numeric weights, so each relationship is shown by its arrow color (increase or decrease) and width (strength) only.
- **Moving relationships.** Drag a relationship's label (or, with labels off, the handle on the selected relationship) to bow the line out of a tangle; double-click to straighten it. Select a relationship and drag either of the two handles at the ends of its arrow onto another concept to reconnect it. While dragging, the concept under the pointer is ringed and the anchor point the drop would use lights up; the gesture commits once, at the drop, as one undo step.
- **Auto Arrange.** No change (Grouped by module was already present in 0.3.0; the manual now lists it).

3.5 Publication-quality figures

- **Figure panel.** Opened with Customize & export on a chart. Figure size (five presets sized for A4 journal pages: single-column portrait 84 by 110 mm, single-column square 84 by 84 mm, double-column landscape 174 by 100 mm, full-page portrait 170 by 240 mm, full-page landscape 240 by 170 mm, or custom, with an aspect-ratio lock); resolution (150, 300, or 600 dpi); format (PNG, SVG, or PDF); plot title, axis titles; four font sizes; legend position (top, bottom, left, right, hidden); palette (default, Okabe-Ito color-blind-safe, grayscale); decimal places; gridlines; line thickness, marker size, opacity; which series are shown and in what order; and, on the effort-gain plane, how many background points to plot. While the panel is open the chart itself is resized to the export size, so the preview is what will be saved.
- **Charts covered.** Baseline Equilibrium convergence plot; Scenarios comparison-to-baseline bar chart and the three trajectory-comparison charts (one shared setting, three files); Transition Point response curves and marginal-return curves; Transition Pathways effort-gain plane, outcome trajectory, and benefit accrual.
- **Exporter.** exportFigure bakes the title and subtitle into the file, so a saved figure says what it shows. PNG files carry their physical resolution through a hand-built pHYs chunk; SVG is the vector chart; PDF is a hand-built single-page file containing the image at its physical size (compressed with the browser's CompressionStream). No new library was added.

3.6 Tables

- **Transition Point results.** Every column sorts ascending or descending on click, numerically where the column is numeric, with a direction indicator; the columns are Intervention, Outcome, TP, Max Change, Range, Efficiency, and Response Type. The sort survives changes of the outcome selection.
- **Sensitivity Analysis.** The impact ranking was a plain list and is now a bordered table (Rank, Concept, Impact) with a sticky header and a scroll box; the elasticity metrics table is aligned the same way. Each has an Export CSV button.

3.7 Transition Pathways

- **Naming.** The gain measure is total outcome gain (TOG) in every label, tooltip, chart axis, CSV and workbook header, and rule text, matching the terminology of the case study of the accompanying paper. The internal variable names follow it, and a pathway setup saved by 0.3.0 with the old beam-search criterion is read correctly.
- **Charts.** The selected pathway is labelled on the effort-gain plane; the outcome trajectory labels the TOG line at each stage; the benefit accrual is a filled area whose shaded region is the Early Benefit Index. A stage waterfall chart was built during this cycle and then removed before release, at the request of the brief.

3.8 The peat meadow example

The illustrative peat meadow case study (food, climate, and biodiversity nexus of Dutch peat meadow polders) is preloaded as 'Peat meadow nexus (Dutch case study)': 20 concepts in five modules (Interventions, Water & soil, Farming system, Nexus outcomes, Socio-economic), 50 relationships, and seven scenarios (the baseline, each of the five levers at full intensity, and a bundle of all five). Weights, starting values, definitions, and positions are those of the case study, so its results can be reproduced. The example loads with a limit of 300 iterations per run because near their transition points the levers settle slowly; other examples are unaffected.

3.9 Branding and the About & documentation tab

- **Logo.** One strand of spaghetti, drawn as an S, runs through the nodes of a causal map, with a cog at its center: the strand is the tangle of causal links, and the cog is the engine that runs it. The mark exists as an inline SVG component (BrandMark) and as a stand-alone SVG file. The favicon is the same mark without the cog teeth, which turn to noise at 16 pixels. PNG lockups for light and dark backgrounds and a 512-pixel mark are rendered from the same drawing, and the mark now heads the cover of the manual, the annex, and this log.
- **Colors and type.** The palette is Cobalt and citrus. Cobalt (#2456D6) is the primary color and navy (#0F1B3D) the dark surfaces; citrus (#FFD23F) is the one accent, with deep citrus (#8A6B00) for yellow text on white, soft citrus (#FFF3BF), cloud (#F6F8FC) as a cool page background, and poppy (#E5484D) for alerts only. Relationships keep their own meaning in the tool (green increases, orange decreases). The tokens are defined once, as a cobalt and a citrus scale in the stylesheet theme and in src/brand.jsx; the former teal of the interface is now cobalt everywhere, and the default chart palette starts with cobalt. The typeface stays Inter.
- **In the interface.** The header shows the mark, the name, and the version, with a citrus rule beneath it; the mark links to the About page. The browser tab has the new favicon, a description, and a theme color. The version shown is read from package.json, so it cannot differ from the release number.
- **About & documentation tab.** A fifth group button, at the right and in citrus, opens one page: a headline over a full-width backdrop of strands, and a description; one box that explains what a fuzzy cognitive map is, why the tool was made (to support participatory processes while providing in-depth analysis), and the reason for the name; who it is for (four groups); where to start; the documentation; and the contact details of Alfred Paarlberg (alfred.paarlberg@wur.nl), with a mail link and a copy button. The footer carries a contact link on every tab. The analysis scope bar is not shown on this tab, and the tool now opens on this page instead of the Model editor.
- **Documentation as PDF.** The manual, the annex, and this log are converted from the Word files by a script (Word file to HTML to PDF with headless Edge, then a contents page with page numbers, bookmarks, and a footer added) and stored in public/docs, so Netlify serves them with the site. src/docs-manifest.json records each file's name, page count, and size, which the page shows. The PDFs must be regenerated at each release, together with the manifest.

4. Implementation Choices

Where the implementation needed a decision, or testing showed that the first choice misled, the choice is recorded here; the Methodological Annex documents the methods.

Topic	Choice in the tool	Reason
Runaway limit	A run without a squashing function stops when any activation passes 1000 in absolute value.	Keeps charts readable and avoids infinite values; the limit is one constant in the source if a different value is preferred.
No clamp without squashing	Activation is not held to -1..+1 in linear mode.	A clamp would be a squashing function in disguise and would hide the dynamics the mode exists to show.
Structural scores in linear mode	Influence and Sensitivity scores use tanh when the setting is linear.	They average absolute responses and need a bounded scale; the panel says so.

Topic	Choice in the tool	Reason
Noise distribution	Uniform, independent per weight, clipped at -1 and +1, rounded to 0.01.	The simplest defensible choice for elicited weights on a bounded scale; weights are stored to two decimals. Clipping pulls the mean of a weight near +-1 slightly inwards.
Summary statistics	Standard deviation over the runs (divisor n); percentiles by nearest rank on the sorted values.	Transparent and reproducible; the tool reports the sorted sample too, through the histograms.
Seeded update order	In asynchronous mode the random order comes from a second seeded stream, the nominal run from a third.	The seed must fix the whole result, and the weight sequence must not depend on the update mode.
Monte Carlo and the workbook	Not part of the Export Analysis Workbook or the figure panel in this release.	Several small distribution charts rather than one figure; a CSV summary exists. Listed under next steps.
Relationship arrows below concepts	Every relationship line sits below every concept box, whatever its state; only the label and handles stay above.	A concept must always be selectable; the invisible click area of a line had been outranking the concept it crossed.
Hover clearing	Leaving a concept clears the highlight after 120 ms; entering sets it at once.	A label overlapping a concept's border made the pointer leave and re-enter several times a second.
Undo history	A drag is recorded in the history at its first movement, not at the mouse press.	A plain click on a relationship label had enabled Undo without changing anything.
Export timing	Figure export waits with a timer, not with requestAnimationFrame.	Animation frames stop in a background tab, which made an export hang.
Documentation as static PDF	The PDFs are files in public/docs, listed in a manifest, not generated in the browser.	No server or library is needed, and the documents shipped with a release always match the version of the tool that links to them.

5. Changes to Existing Features

- **Tab bar:** ten tabs in four groups instead of a single row of nine, and the About & documentation button at the right; the tool opens on the About page. The header and footer gain the new logo, the version, and a contact link.
- **Simulation:** a fourth transfer function; `simulate()` takes an optional random generator for the asynchronous order; a run that reaches its iteration cap now reports the cap (it reported one more).
- **Replacing a model:** starting a blank model, loading an example, importing a file, and building from a matrix now also clear the Monte Carlo and Transition Pathways results (they were kept, marked out of date).
- **Transition Pathways:** the gain measure is TOG throughout; the diagnostics note that a runaway counts as not converged; a notice appears when pathways fail to converge without squashing.
- **Sensitivity Analysis and Transition Point Analysis:** a note appears while no squashing function is selected.

- **Exports:** charts export through the figure exporter (PNG, SVG, PDF) with the title in the file; the Analysis Workbook records the transfer function.
- **Trajectory charts:** the Baseline Equilibrium convergence plot and the baseline, scenario, and difference trajectories of Scenarios & Simulation are now drawn as smooth lines without a marker at every iteration; the marker size in the figure panel can still add them, and Reset appearance returns to the marker-free default.
- **Version** raised from 0.3.0 to 0.4.0 in package.json and package-lock.json. No other analytical method, default setting, or existing export has changed.

6. Defects Found and Fixed During Development

Problem	Cause	Fix
The Monte Carlo tab crashed when opened.	It used the chart-cache hook with a key that is not registered in the chart registry.	The hook is not used by this tab (Section 4).
A figure export never finished in automated tests.	requestAnimationFrame does not fire in a background tab.	A timer instead.
Exported figures came out larger than the requested size.	The title band and padding were added around the chart.	The chart is shrunk by exactly the reserved band before capture; sizes now match to within a pixel.
Exported PNG files opened at the wrong physical size.	The canvas encoder records no resolution.	A pHYs chunk carrying the chosen dpi.
Concepts were hard to select and flickered on hover.	Relationship lines and their click areas sat above concept boxes; labels overlapping a border caused repeated enter and leave events.	Lines below all concepts; hover clearing debounced by 120 ms; opacity transitions.
Reconnecting a relationship was nearly impossible.	The library's own handle sat up to 67 pixels from where the arrow is drawn.	Custom handles at the drawn ends, with a ring and a lit anchor on the target.
Dragging was slow.	Every mouse move wrote the model.	A local preview during the drag; one model change at the drop; frame batching.
A reconnect drop missed its target.	The relationship's own label followed the pointer and covered the target.	Label and handles ignore the pointer during the drag.
Undo became available after only clicking a relationship.	The history was recorded at the mouse press.	Recorded at the first movement.
Double-clicking a bend handle added a concept.	The handle layer was not excluded from the add-concept double-click.	Excluded.
Collapsing the Inspector stacked the panels.	A one-column grid wraps a second item.	The Inspector is not rendered while collapsed.

Problem	Cause	Fix
A search feature would have failed at start.	It used a value declared later in the same function.	Moved after the declaration (caught before release).
Stale results stayed after replacing a model.	Monte Carlo and pathway results were not cleared by the three replace paths.	Cleared with the others.
The same seed did not repeat an asynchronous Monte Carlo run.	The update order used the unseeded generator.	Seeded order stream; two new tests.
A run at its cap reported one iteration too many.	The loop counter ends one past the cap.	Reported as the minimum of the two.
Runaway runs went unremarked on Sensitivity, Transition Point, and Pathways.	Only Baseline and Scenarios knew about divergence.	A note on the sweep tabs, a divergence count on Monte Carlo, and a notice on Pathways.
A Monte Carlo run went on after leaving the tab.	Nothing cancelled it on unmount.	Cancelled on unmount.

7. Verification

- **Build and static checks:** the production build (vite build) completes without errors; Babel-based static checks over the whole source found 0 rules-of-hooks violations, 0 undefined identifiers, and 0 unused top-level declarations.
- **Engine tests in Node,** extracted from the app source (119 pass): the six earlier suites unchanged (pathway engine 22, analysis functions 21, feedback loops 5, Model check 9, folding 12, example model 11); Monte Carlo (20): zero spread reproduces the nominal result, real variance appears, the same seed repeats and another differs (also in asynchronous mode), the statistics match independent calculations, pinned weights and starting values are never perturbed, progress is reported, runaway runs are counted; linear mode (19): the identity update, no clamp, runaway stop and flag, damped and absolute-rule cases, bounded functions unchanged, the pathway engine equal to simulate() under linear, and the iteration cap.
- **In the running app:** every tab opened on all four example models, including the blank one, without a display error; every analysis run under each transfer function, both update rules, and both update modes (seven combinations) without errors; the model, including the transfer function, survived a reload and an Export JSON and Import round trip; the Analysis Workbook exported under linear; drags, hover, search, and undo checked on the 40-concept example; the tab groups checked at phone width.
- **About page:** checked in the browser: the three download links and the open-in-a-new-tab links, the mail link and the copy button, the buttons that jump to other tabs, and the tab groups at desktop and phone widths; the scope bar is hidden on this tab.
- **Peat meadow example:** the baseline reproduces the case study (converges in 16 iterations), all seven scenarios converge, and the pathway analysis reproduces its pathway table (188 pathways, 14 admissible, with the archetypes, TOG, CII, PEI, and IG of the case study to three decimals).
- **Monte Carlo on the examples** (1,000 runs, baseline, seed 2026): on the peat meadow model, weights varied by ± 0.1 give 1,000 of 1,000 runs that settle and outcomes that barely move (the baseline is almost saturated); at

+ -0.25, 915 of 1,000 settle, and biodiversity, at -0.99 nominally, has a 95th percentile of +0.55, so some runs end in another state. On the water-energy-food example only 135 of 500 runs settle within 100 iterations at + -0.1 (236 within 500), although its baseline settles in 73: the model lies close to an oscillation boundary.

8. Deployment Status

- **Not yet deployed.** The 0.4.0 changes exist only in the local working copy (src/SpaghettiEngine.jsx, about 2,400 lines added and 320 removed against 0.3.0, and the version number in package.json and package-lock.json). Nothing has been pushed; 0.3.0 remains live. When the release is pushed to main, Netlify rebuilds automatically, and the commit and the live bundle should be recorded here.
- **PDF files:** the three documents add PDF files of about 2.8 MB in total to the deployment (public/docs). The logo source files are kept in brand-assets, outside the deployed site; only the favicon and the touch icon are public.
- **Build note:** npm run build fails in the local OneDrive folder because npm's command shim cannot handle the ampersand in the folder path; running node node_modules/vite/bin/vite.js build directly works. The local build is one JavaScript file of about 3.7 MB and a 60 KB stylesheet.

9. Prototype Scope and Known Limitations

- **Monte Carlo** perturbs weights and starting values only, independently and uniformly, with one spread for all; it does not sample the structure, correlations between weights, or per-relationship uncertainty. Its results are not saved, not in the Analysis Workbook, and not covered by the figure panel; its settings reset when you leave the tab.
- **Runs that do not settle** are still included in the Monte Carlo statistics and flagged. In models that converge slowly or lie near an oscillation, perturbed runs can exceed the iteration limit even though the nominal run does not; raising the limit helps only in part.
- **Without a squashing function** most maps run away, and the runaway limit of 1000 makes the stopping point, not a real end state, what is reported. Sweeps that include such runs say little about the model, which the tabs state.
- **Network:** connection anchors are snapped to the nearest of twelve points per concept; creating a new relationship by dragging from a dot was verified only in part in automated tests, because the library tracks it with pointer events.
- **Documentation PDFs** are static and show the release they were built for; they are converted from the Word files by a script whose layout is close to, but not identical to, the Word layout (the contents page numbers are computed from the PDF itself).
- **Interface details:** numeric fields snap back to 0 when cleared, so a value is overwritten rather than deleted; the header bar is not built for phone widths.
- **Not yet implemented** from the interface brief: the influence, impact, intervention, and outcome view modes of the Network tab, richer plain-language tooltips, and a basic and advanced mode.
- **Unchanged:** pathway results are not saved; exhaustive pathway construction is capped at 20,000; native PDF, DOCX, and PPTX export of a whole analysis is still on the roadmap; the bundled JavaScript exceeds Vite's 500 KB advisory threshold, and the npm audit findings documented for 0.1.0 and the placeholder favicon are unchanged.

10. Related Documentation

Document	Covers
The Spaghetti Engine - User Manual (version 0.4.0)	How to use every feature, now including the Monte Carlo tab (Section 11), the tab groups (Section 3.1), the option to run without a squashing function (Sections 2.2, 8.1, 9.5), the reworked Network tab (Section 4.1), and the figure panel (Section 14.2).
The Spaghetti Engine - Methodological Annex (version 0.4.0)	The methods, now including the Monte Carlo simulation (Section 11), the linear update and runaway rule (Sections 3.3 and 3.4), and the total outcome gain (Section 13).
Transition Pathway Analysis - methods extension (draft)	The methodological framework the Transition Pathways tab implements, now using the term total outcome gain.
Spaghetti_Engine_Dev_Log_v0.3.0	The 0.3.0 release: the Transition Pathways tab.
This Development Log	What changed in 0.4.0 and the technical work behind it.

11. Next Steps

- Apply the Monte Carlo engine to the pathway analysis: repeat it over sampled weight matrices, so that rank stability of the pathways becomes a measure of confidence (the roadmap item noted in 0.3.0).
- Extend Monte Carlo with per-relationship uncertainty ranges, an inclusion in the Analysis Workbook, and the figure panel; consider variance-based sensitivity indices on the same runs.
- Bring the methods extension in line with the choices of the 0.3.0 log (iteration cap, sweep-resolution EBI, invariance tolerance).
- Complete the Network brief: view modes, plain-language tooltips, and a basic and advanced mode; test new-relationship dragging with a real pointer.
- Move the pathway and Monte Carlo analyses into a Web Worker; code-splitting to reduce the initial download; replace the placeholder favicon; native PDF, DOCX, and PPTX export.